

Srinjoy Bhuiya

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EDUCATION

- **University of Alberta** Edmonton, CA
Master's in Computing Science (Thesis) GPA: 3.9/4 Sep 2023 – May 2026
 - **Courses:** Reinforcement Learning 1 & 2, ML & the Brain, Earth Observation AI Applications
 - **Teaching Assistant:** CMPUT328 (Visual Recognition), CMPUT200 (Ethics of Data Science & AI)
 - **Supervised by Dr. Nilanjan Ray at the VISUAL Laboratory:**
- **Heritage Institute of Technology** Kolkata, India
Bachelor of Technology in Computer Science; CGPA: 9.34/10 Aug 2019 – May 2023
 - **Courses:** Analysis of Algorithms, Pattern Recognition, AI, Machine Learning, NLP:

HONORS AND AWARDS

- **Graduate Recruitment Scholarship** Edmonton, Canada
University of Alberta Sep 2023 – Sep 2024
 - **Grant:** Awarded to high-achieving graduate students with significant research potential at the University of Alberta.
- **Central Scholarship for College and University Students** India
Government of India Apr 2019 – Mar 2023
 - **Grant:** Awarded to the top 80,000 students nationwide for exemplary performance in Higher Secondary examinations.

EXPERIENCE

- **Synopsis Medical Technologies** Edmonton, Canada
Machine Learning Research Scientist (Part-Time) May 2025 – Sept 2025
Research Intern (MITACS Accelerate Fellowship) May 2024 – Apr 2025
 - **ML Research Scientist:** Involved in the design and development of a Medical OCR and document triage system using quantized SLMs for medical document parsing and understanding, speeding up medical administrative workflows
 - **Mitacs Accelerate Fellowship:** Developed a novel diffusion based anomaly detection method to classify cardiovascular pressure signals ensuring high-quality data selection for related downstream tasks.
- **Indian Statistical Institute (Center for Soft Computing Research)** Kolkata, India
Research Intern (Part-Time) Jan 2023 - Sept 2023
 - **Edge Detection on Ultra-high Resolution Images:** Working under Dr Kuntal Ghosh, developing a novel deep-learning model for understanding the rate of forest cover reduction in the Kaziranga Forest National Park, Assam, India using multi-sensor satellite data.
- **Laboratory of Translational Medicine - Toronto Metropolitan University** Ontario, Canada
Research Intern (Full-time, MITACS Globalink Research Program) Jul 2022 - Oct 2022
 - **Catheter Segmentation:** Developed a novel real-time segmentation model for catheter tracking on volumetric ultrasound images using PyTorch.
 - **Catheter Tracking in 3D Ultrasound Images:** Developed a 3D RANSAC algorithm with polynomial curve fitting regression to generate catheter paths in ultrasound images of the prostate.
- **Future of Work Lab - University of California, Berkeley** Remote
Research Analyst Intern (Full-time, Contractual) Jun 2022 - Jul 2022
 - **Data Preprocessing:** Conducted thorough digitization of physical census data for further analysis.
 - **Research Data Analysis:** Applied PCA and various statistical methods to analyze the underlying relations between literacy rates and employability.
- **Ajantrik Private Limited** Kolkata, India
Technical Research Associate (Full-time) Dec 2021 - Jun 2022
 - **Collision Avoidance Systems:** Developed collision avoidance perception software for warehouse-based Autonomous Guided Vehicles(AGVs).
 - **Real-Time Path-finding system:** Developed approximation-based path-finding algorithms for warehouse AGVs which helped reduce the time to collect orders by 30% .

- **Future Ready Talent Program(In association with AICTE and Microsoft)** Remote
Sep 2021 - Dec 2021
Virtual Internship (Part-time)
 - **Learning Microsoft Azure:** Learned the in-demand azure cloud & security skills aligned to industry needs.
 - **Cloud Based Biomedical Computer Vision project:** Developed a Cloud-Computing based project to detect the presence of glaucoma using a smartphone based pictures of the eye.

RESEARCH & PUBLICATIONS

- **Improving Stream-flow Prediction in Snow-Dominated Regions using Pretrained Fusion Attention Models:**
(Sep 2025 - Present)
Working under the guidance of Dr Nilanjan ray at the University of Alberta to develop a long-term(seasonal) river water-flow forecasting in data-scarce & snow-dominated river basins using large scale pretraining on a novel attention-fusion architecture along with specialized multi-stage finetuning for the Bow river basin near Calgary.
Authors: 1. Srinjoy Bhuiya 2. Nilanjan Ray
- **High-Quality 3D Biomedical Image Generation using Generative Adversarial Networks (Accepted and presented at PREMI,2023. Published in Springer Lecture Notes in Computer Science):**
(Oct 2022 - May 2023)
Working under the guidance of Dr Dinabandhu Bhandari at Heritage Institute of Technology to create a novel generative model for generating high-resolution 3D medical images for image augmentation.
Authors: 1. Srinjoy Bhuiya, 2. Subhopriyo Sadhukhan, 3. Suchandra Chakraborty, 5. Dr Dinabandhu Bhandari 6. Dr. Deba Prasad Mandal
Link: Generation of Data for Training Retinal Image Segmentation Models
- **Using attention-based auto-encoders for catheter path reconstruction in ultrasound images (Accepted at SPIE Medical Imaging):**
(Peer-Reviewed)
(Jul 2022 – Oct 2022)
Worked with Dr. Elodie Lugez at Toronto Metropolitan University to create a novel architecture for tracking and localization of Catheters in 3D ultrasound images of the prostate.
Authors: 1. Srinjoy Bhuiya, 2.Shreyasi Mandal 3. Dr Elodie Lugez
Link: <https://doi.org/10.1117/12.3003897>
- **Retinal Blood vessel segmentation from Fundus Images(Accepted and Presented at IEEE CALCON 2022):**
(Peer-Reviewed)
(Apr 2021 – Jul 2021)
Worked with Dr Anindya Sen at Heritage Institute of Technology, Kolkata to create a novel architecture for efficient blood vessel segmentation of colour fundus images. I am scheduled to present the paper at IEEE CALCON on 10th December.
Authors: 1. Srinjoy Bhuiya, 2. Soumik Roy Choudhury, 3. Geetanjali Aich, 4.Muskan Maurya, 5. Dr Anindya Sen
Link: <https://ieeexplore.ieee.org/abstract/document/10060189>
- **Exploring the Effects of Data Augmentation for Drivable Area Segmentation:** (Nov 2021)
Conducted research dealing with the various weather scenarios that an autonomous vehicle encounters. We proposed a novel data augmentation and driving area segmentation model to achieve real-time detection in various challenging conditions.
Authors: 1. Srinjoy Bhuiya, 2, Ayushman Kumar, 3. Sankalok Sen
Arxiv Preprint: <https://arxiv.org/abs/2208.03437>

SKILLS SUMMARY

- **Generative AI & LLMs:** Transformers, RAG, LangChain, Fine-tuning (LoRA, PEFT), Prompt Engineering, Vector DBs, RLHF
- **Deep Learning:** TensorFlow, PyTorch, Hugging-Face, Computer Vision (ViT, YOLO &, CNNs)
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, Statistical Modeling, NLTK
- **Data & Visualization:** SQL, Pandas, Tensor-board, NumPy, Matplotlib, Plotly, Weights & Biases (W&B)
- **Cloud & MLOps:** AWS , GCP , Azure, Docker, Kubernetes, CI/CD for ML
- **Software Development:** C, C++, Python ,Linux, Git, JAVA, REST APIs (FastAPI, Flask)

VOLUNTEER EXPERIENCE

- **Director of Finances, CSGSA** Edmonton, CA
Sep 2023 - Apr 2025
Executive member of the Computing Science Graduate Students Association at the UofAlberta
- **Technical Lead at Developer Student Clubs HITK** Kolkata, India
Sep 2020 - Sep 2023
Conducted several technical & soft-skills training sessions impacting hundreds of students.
- **Head Coordinator of the HITK Quiz Club** Kolkata, India
Apr 2021 - May 2022
Conducted several technical, popular culture, and general quizzing sessions.